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Title: Higher methanotroph abundance and bottom-water methane in ponds with floating photovoltaic arrays

Running head: Floating solar and methane in ponds

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Author Contribution Statement:

NER designed the study, provided supplies, performed field sampling, laboratory analyses, and data analysis, and wrote the first draft of the manuscript. SMG secured funding, constructed the floating solar arrays, performed field sampling, and edited the manuscript. AC, SJC, and MT performed field sampling and edited the manuscript. MAH secured funding, performed field sampling, and edited the manuscript. SA performed field sampling, laboratory analyses, data analysis, and edited the manuscript. MLS designed the study, provided supplies, data analysis, secured funding, and edited the manuscript.

Scientific Significance Statement (125 word max)

Producing energy using floating photovoltaic (FPV) powerplants offers an opportunity to produce renewable energy, spare land, and reduce evaporation from ponds, lakes, and reservoirs. However, FPV deployment in these ecosystems is associated with colder temperatures, less oxygen availability, and changes in carbon cycling processes. Specifically, experimental evidence demonstrates an increase in concentrations of methane – a potent greenhouse gas – in ponds following FPV deployment. In this study, we investigate how microbial communities associated with aquatic methane cycling differ between ponds with and without FPV. We show that FPV deployment increases bottom-water methane concentrations and triggers dense blooms of methane-oxidizing bacteria. These results provide the first evidence that microbial communities respond strongly to engineered shading and may help buffer greenhouse gas emissions in solar-covered waters.

Data and Code Availability

All raw and processed data for this project are publicly available. The code used for statistical comparisons, generating figures, and processing microbial community data are available on GitHub at https://github.com/MarschmiLab/Ray_LO_Letters_FPV_Methane. Biogeochemical data and summaries of methane cycling microbe abundances is available for download via the Figshare Repository (<https://doi.org/10.6084/m9.figshare.29614025.v1>). The raw, compressed 16S rRNA gene sequencing fastq files are available in the NCBI Sequence Read Archive (SRA) under the BioProject accession number PRJNAXXXXXXX (This will be made available upon acceptance or by reviewer request). All flow cytometry data are available in the Flow Repository database with the ID number XXXXX (This will be made available upon acceptance or by reviewer request).

Abstract (150 words max)

Floating photovoltaic (FPV) arrays alter the biogeochemistry of waterbodies on which they are deployed. Here, we investigated dissolved methane (CH₄) dynamics and associated CH₄ cycling microbial communities in the second-year of FPV deployment (70% aerial coverage) in experimental ponds. We found that bottom-water CH₄ concentrations were twice as high in ponds with FPV compared to those without. Surface water CH₄ concentrations were orders of magnitude lower than bottom-waters, but were similar between ponds with and without FPV. There were no differences in potential sediment CH₄ production or CH₄ associated microbes. Instead, dissolved CH₄ patterns could be explained by elevated abundance of methanotrophic bacteria in the water column of FPV ponds, where late-summer abundances exceeded 1,000,000 cells mL⁻¹. Prolonged periods of CH₄ production in low oxygen FPV ponds likely favored blooms of methanotrophs in surface waters where some oxygen is available, mitigating diffusive CH₄ emissions to the atmosphere following FPV deployment.

Keywords: pond, methane, floating solar, greenhouse gas, methanogen, methanotroph

Introduction

Methane (CH₄) emissions from aquatic ecosystems have increased because of human activity (Jackson et al. 2024). Floating photovoltaic arrays (FPV), first deployed in 2007, are a rapidly expanding renewable energy technology that can alter aquatic CH₄ dynamics and emissions (Cazzaniga and Rosa-Clot 2021; Ray et al. 2024). By 2023, over 643 FPV power plants had been installed on lakes, ponds, and reservoirs worldwide (Nobre et al. 2024; Ramanan et al. 2024). There is substantial interest and potential in expanding FPV use on reservoirs and lakes to meet demands for renewable energy (Jin et al. 2023; Woolway et al. 2024), but the ecological and biogeochemical tradeoffs – particularly effects on CH₄ cycling – must be evaluated before broader deployment (Almeida et al. 2022; Gallaher et al. 2025).

FPV deployment is associated with lower water temperatures (Dörenkämper et al. 2021; Andini et al. 2022), reduced oxygen availability (Wang et al. 2022; Ray et al. 2024), and changes in stratification (Ilgen et al. 2023). These factors influence CH₄ dynamics in small, shallow systems where FPV has most commonly been deployed (Bastviken et al. 2004; Aben et al. 2017; Ray and Holgersson 2023; Nobre et al. 2024). Indeed, in the first year of FPV deployment, experimental ponds with FPV showed twice the dissolved CH₄ concentrations and CH₄ ebullition compared to control ponds, but diffusive CH₄ emissions were lower (Ray et al. 2024). Whether these observed patterns persist over time or might be similar in other ecosystems is unclear.

Understanding how FPV alters CH₄ cycling microbes may reduce uncertainty in CH₄ emission responses. Methanogenic archaea (CH₄ producers) and methanotrophic bacteria (CH₄ consumers) regulate CH₄ fluxes, but it remains unresolved how FPV-induced changes in

temperature, light, oxygen, and primary production affect their abundance, composition, and activity. Evidence from non-FPV settings suggests these conditions will re-structure CH₄ cycler communities and alter metabolic rates (Wang et al. 2021; Bertolet et al. 2022). Clarifying microbial responses to FPV is critical for predicting whether this technology amplifies or dampens CH₄ emissions and, more broadly, can inform how pond microbial communities respond to reduced light and oxygen.

Here, we investigate how FPV affects CH₄ dynamics in temperate experimental ponds during the second year of FPV deployment. We pair observations of temperature, dissolved oxygen, dissolved CH₄, and the CH₄-associated microbial community throughout the summer with CH₄ process rates during an intensive mid-summer sampling campaign in ponds with and without FPV. Because methanotrophs can be challenging to resolve via 16S alone, we focus on well-characterized CH₄ oxidizers and pair community data with functional rate measurements to capture both composition and activity.

Methods

Location and Sampling Scheme

In 2023, FPV arrays (70% aerial coverage) were deployed on three ponds at the Cornell Experimental Ponds Facility in Ithaca, New York, USA (42.5049 N, 76.4666 W). The experiment used a before-after-control-impact design, with observational sampling of six ponds starting in 2022 with FPV deployment in 2023 (Ray et al. 2024). The ponds are 30 x 30 m inverted, truncated pyramids, 1.75 m deep, unlined, with sediment accumulation since their construction in 1958-1959. Rooted, submerged macrophytes dominate primary production.

We returned in summer 2024 to continue measurements of dissolved CH₄ (surface and bottom), temperature, and dissolved oxygen (4 depths; 0.1, 0.75, 1, and 1.25 m from the water surface). We sampled CH₄-cycling microbes four times in 2024 from surface water, bottom water, and sediments. On July 11, 2024, we measured diffusive CH₄ exchange and net CH₄ production or consumption in surface waters, bottom waters, and sediments.

Observational CH₄ Concentration Measurements

We collected samples to determine dissolved CH₄ concentrations in the surface and bottom water of each pond six times between June 20 and September 5. Samples were collected in triplicate from the pond center using a headspace equilibration approach, with surface samples collected by hand at 10 cm depth and bottom water collected using a Van-Dorn bottle ~10 cm from the sediment-water interface (Ray et al. 2024). Samples were stored in 12 mL pre-evacuated borosilicate exetainers until analysis using a gas chromatograph equipped with a flame ionization detector (Shimadzu GC-2014).

Microbial Sample Collection and Preservation

Water samples for microbial analysis were collected as for CH₄ measurements and sequentially filtered through 200 µm and 20 µm NYTEX mesh into 2 L Nalgene bottles and kept on ice until return to the lab the same day. Surface sediments (top 2 cm) were collected using a PVC pole corer and stored in WhirlPak bags in the cooler. Water was filtered onto 0.22 µm polyethersulfone filters using a peristaltic pump (190-2020 mL per sample). Sediments were homogenized using a stainless-steel coffee grinder (three 1 second pulses per sample). Filters

(halved) and homogenized sediment were stored in 2 mL cryovials at -80 °C. Samples were fixed with 1 µL 25% glutaraldehyde, diluted 15–20×, stained with SYBR Green I, and run in triplicate on an Attune NxT (*see supplemental methods*).

Microbial DNA Extraction, Sequencing and Bioinformatics

DNA was extracted from half filters using the DNeasy PowerWater Kit and from sediments using the DNeasy PowerSoil Pro Kit (Qiagen), eluted in 50 µL elution buffer, quantified with a Qubit fluorometer (Thermo–Fisher) and stored at -20 °C. The V4 region of the 16S rRNA gene was amplified following the Earth Microbiome Project protocols (Caporaso et al. 2011; Apprill et al. 2015; Parada et al. 2016) using triplicate 25 µL PCRs with KAPA HiFi 2X Mastermix (Roche), 0.2 µM 515F (5'-GTGYCAGCMGCCGCGGTAA) and 806R (5'-GGACTACNVGGGTWTCTAAT) primers with Illumina Nextera overhangs, and 5 ng template DNA. Zymo-BIOMICS Microbial Community DNA Standard (Zymo) was included to assess error rates. Samples, including PCR and indexing blanks, were pooled, indexed, and sequenced (2 x 250 bp) on an Illumina MiSeq sequencer at the Cornell Biotechnology Resources Center.

Raw sequences were processed into amplicon sequence variants (ASVs) using the DADA2 workflow (Callahan et al. 2016) in R. Reads were trimmed to 253-254 base pairs (max error = 1), merged, denoised, and chimera-checked. Taxonomy was assigned with GreenGenes2 (McDonald et al. 2024). Data were imported into *phyloseq* (McMurdie and Holmes 2013); mitochondrial, chloroplast, and ASVs with a higher relative abundance in blanks were removed. The mock community revealed 27 spurious ASVs (error rate of 0.099%). Samples with low read counts were excluded, resulting in 8,188-171,275 reads per sample. Water column ASV relative

abundances were multiplied by cell counts to obtain absolute ASV counts per mL whereas the ASV relative abundances in sediments were multiplied by the minimum library size (20,822 sequences) for relative abundance.

Microbial metabolic profiles were predicted using Functional Annotation of Prokaryotic Taxa (FAPROTAX) based on cultured prokaryotes (Louca et al. 2016). Methanogens and methanotrophs were identified at the order level and verified through literature and public databases (NCBI; Table S6).

Diffusive CH₄ Flux Measurements and Estimation of k₆₀₀

We measured diffusive fluxes at the center and edge of each pond using floating chambers (18.93 L volume; 0.071 m² cross sectional area) attached to a portable analyzer that determines CH₄ concentrations in air using off-axis integrated cavity output spectroscopy (GLA132-GGA, ABB Measurement and Analytics; Ray et al. 2024). At the same time, we collected headspace samples for determination of dissolved CH₄ concentrations. Using diffusive CH₄ flux measurements and dissolved CH₄ concentrations, we calculated gas transfer velocities.

Water Column CH₄ Bottle Incubations

We used light and dark bottle incubations to determine rates of CH₄ production and consumption in surface and bottom waters of each pond. Immediately following the collection of dissolved CH₄ samples, we filled 125 mL glass incubation bottles (one clear and one darkened with electrical tape) and sealed them with a gas tight rubber septa. These bottles were then suspended either 10 cm from the surface or 10 cm from the sediment-water interface. The following day,

bottles were collected, de-capped, and sampled for temperature, dissolved oxygen, and dissolved CH₄ concentrations collected as described previously. Rates of CH₄ production or consumption were then calculated as the change in dissolved CH₄ concentration over time, assuming starting conditions in each bottle matched the conditions in the pond when the bottles were filled.

Sediment CH₄ Bottle Incubations

We collected sediments alongside microbial samples and stored in acid-washed polyethylene bottles at -20 °C. To determine rates of potential sediment CH₄ production, we performed bench-top bottle incubations. We added ~ 5 g wet weight of thawed sediment to 125 mL glass incubation bottles equipped with optical O₂ sensor spots (FireSting -GO2, PyroScience GMBH), added 70 mL of tap water that had been left out for > 24 hrs, sealed the vials with gas tight rubber septa, and then over-pressured each vial with 15 mL of lab air. Two replicate bottles were used per pond. Samples for analysis of headspace CH₄ concentration were collected on day 0, 2, 5, 6, and 14 by collecting 3ml headspace in 2.5 mL gas-tight vials, analyzed as described previously. Incubations were performed in the dark with room temperature ~22 °C. To compare sediment CH₄ production potential, we calculated rates of sediment CH₄ production over time relative to the dry mass of sediment in each bottle.

Statistical Comparisons

All statistical analyses were performed using R statistical software and figures were made using the *ggplot2* package (Wickham 2016). We considered the results of statistical tests to be significant when $p < 0.05$.

We fit linear mixed-effects models (fixed: treatment, day of year; random: pond) to compare environmental and microbial metrics in FPV compared to control ponds using *lme4* (Bates et al. 2015; Table S1). Rates of water column CH₄ production or consumption in surface and bottom waters were compared using a mixed effects model of light or dark bottle and FPV presence with pond as a random effect (Table S2). Contrasts between treatments were made using least square means tests via *emmeans* (Lenth 2018) (Table S3).

Water temperature and oxygen were depth-averaged by pond and day. Diffusive CH₄ fluxes, k₆₀₀, and potential sediment CH₄ production were compared using two-tailed t-tests (Tables S4–S5).

Bray–Curtis dissimilarities were visualized with PCoA and tested using PERMANOVA (adonis2) and β-dispersion (betadisper) in *vegan* (Oksanen et al. 2018). Water column ASVs were scaled to total cell counts; sediment ASVs were rarefied. Differentially abundant ASVs were identified with ANCOMBC-II (tax_level = “ASV”; prevalence >10%, hol-adjusted q < 0.05, global test by FPV; Lin and Peddada 2024).

Results

FPV Ponds were Colder and Less Oxygenated

Ponds with FPV were on average almost 4 °C colder across the water column (18.87 ± 1.69 °C; mean ± SD) than control ponds (22.78 ± 2.29 °C) during summer ($p < 0.001$; Fig. S1). While all ponds were generally undersaturated in DO, surface waters in control ponds were occasionally oversaturated, unlike FPV ponds, which remained undersaturated throughout (Fig. S2). Mean

DO in FPV ponds (30.18 ± 10.59 %) was less than half that of control ponds (62.34 ± 10.59 %; $p < 0.001$).

FPV Ponds had Higher CH₄ Concentrations at Depth

Dissolved CH₄ concentrations were relatively stable over time (Fig. 3A & 3C). Surface CH₄ did not differ between FPV (1.79 ± 4.01 $\mu\text{mol CH}_4 \text{ L}^{-1}$) and control ponds (2.07 ± 3.19 $\mu\text{mol CH}_4 \text{ L}^{-1}$; $p = 0.208$; Fig. 3B). However, bottom water CH₄ was two orders of magnitude higher than in surface waters, and more than twice as high in FPV ponds (306.2 ± 419.0 $\mu\text{mol CH}_4 \text{ L}^{-1}$) than controls (129.7 ± 45.60 $\mu\text{mol CH}_4 \text{ L}^{-1}$; $p = 0.003$; Fig. 3D).

FPV Increased Late-Summer Methanotroph Abundance in the Water Column

CH₄ associated microbial abundance differed between FPV and control ponds, driven primarily by increases in water column methanotrophs in late-summer. In surface waters, methanotroph abundance increased throughout summer and was five times higher in FPV ponds ($4.6 \pm 4.6 \times 10^5$ cells mL^{-1}) than controls ($0.9 \pm 0.9 \times 10^5$ cells mL^{-1} ; $p < 0.001$; Fig. 2A-B). Methanogen abundance was orders of magnitude lower and showed no surface water differences between treatments (Fig. 2C & D). Bottom waters displayed similar trends, with 3.5 times more methanotrophs in FPV ponds (Fig. 2E-H). In contrast, sediment methanogens averaged ~20% of the community and were more abundant than methanotrophs, (~5%), but neither group differed between treatments (Fig. 2I-L).

FPV Altered CH₄-Cycling Communities

FPV significantly altered CH₄-cycling communities in both water and sediment (Fig. 3; Table 1). In the water column, FPV had the strongest effect (pseudo-F = 8.20; R² = 0.11), while pond explained slightly more variance (R² = 0.13) but caused less separation (pseudo-F = 2.40). Day of year also contributed to variation (R² = 0.10). In sediments, pond was the dominant driver of community change (pseudo-F = 4.53; R² = 0.24), but FPV still induced clear shifts (pseudo-F = 8.41; R² = 0.11), with additional variance explained by day of year (R² = 0.08).

Taxon-specific analyses revealed similar patterns (Fig.S3A&B; Table S7): pond identity dominated methanogen community structure (R² = 0.28), while FPV exerted stronger effects on methanotrophs (pseudo-F = 7.69; R² = 0.10) than on methanogens (pseudo-F = 10.47; R² = 0.12) (Table S7). FPV effects were most pronounced in the water column, likely driven by methanotrophs given the absence of methanogens, while sediment communities reflected stronger spatial structuring.

Type I Methanotrophs Drove CH₄ Cycler Community Change

Four methanotrophic ASVs were higher in the water column of FPV ponds compared to controls: from Methylococcaceae (*Methyloparacoccus*) and Methylomondadaceae (*Methylobacter_C*, *Methylococcales*, *Methylomonas*) (Fig. 4). *Methyloparacoccus* reached densities over 500,000 cells/mL (p < 0.001). The only enriched methanogen, *Methanobacterium_B*, was present at lower abundances (<10,000 cells/mL; p = 0.037). Only one differentially abundant ASV was detected in the sediments, *Methanoperedens_A* (ASV_4603; a methanogen), but its relative abundance was negligible.

No Apparent Differences in CH₄ Cycling Processes

We found no differences in diffusive water-air CH₄ exchange or gas transfer velocities between ponds with and without FPV or between the center and edge of ponds (Table S4). On average, all ponds emitted CH₄ to the atmosphere at a rate of $404.1 \pm 896.6 \mu\text{mol CH}_4 \text{ m}^{-2} \text{ h}^{-1}$. We also found no difference in rates of water column CH₄ net-production or consumption between ponds with and without FPV (Table S3). Sediments from both open and FPV ponds produced CH₄, but we found no difference in potential CH₄ production by sediments in ponds with FPV ($5.43 \pm 4.81 \text{ ppm CH}_4 \text{ g dry sediment}^{-1} \text{ d}^{-1}$) compared to those without ($14.13 \pm 10.92 \text{ ppm CH}_4 \text{ g dry sediment}^{-1} \text{ d}^{-1}$; $p = 0.303$; Table S5).

Discussion

We found that dissolved CH₄ concentrations remain elevated in bottom waters for two years following FPV deployment (Fig. 1D) and water column methanotroph abundance is enhanced in ponds with FPV (Fig. 2). Surface water CH₄ – and diffusive CH₄ emissions – were similar between FPV and control ponds. Taken together, this decoupling suggests that FPV induced changes to redox conditions allows for a buildup of CH₄ in bottom-waters, but biological responses limit CH₄ flux to the atmosphere.

Specifically, we hypothesize that lower oxygen conditions in FPV ponds prolong anoxic conditions, allowing for greater CH₄ accumulation at depth. These shifted redox conditions allow for sustained CH₄ production in sediments and accumulation of CH₄ in waters overlying the sediments. A persistently large CH₄ pool likely fuels type I methanotrophy in the water column where oxygen is more available. Methanotrophs reached bloom-like densities in late summer,

exceeding one million cells per milliliter and reaching levels ten times higher in FPV than controls (Fig 2A). By intercepting CH₄ near the air-water interface, methanotrophs likely reduce CH₄ concentrations before atmospheric release.

Beyond CH₄ availability, changes in microbial community composition also likely reflect shifts in redox conditions under FPV-driven hypoxia. For example, several *Methylomonadaceae* taxa were more abundant in FPV ponds. These type I CH₄-oxidizing bacteria are adapted to low-oxygen environments and high CH₄ availability (Li et al. 2024). Type I methanotrophs use the ribulose monophosphate (RuMP) pathway for carbon assimilation, which is more energy efficient than the serine pathway used by Type II methanotrophs and supports faster growth rates under high CH₄ conditions (Knief, 2015). Notably, methanotrophs are rarely measured in freshwater CH₄ studies (Wang et al., 2021) and our results highlight the importance of examining both CH₄ producers and consumers to understand how engineered systems influence microbial CH₄ cycling. Whole-community perspectives also reveal how broader microbial assemblages mediate CH₄ production responses (Bertolet et al. 2022).

By pairing amplicon sequencing with flow cytometry, we move beyond relative abundance to quantify absolute abundance of CH₄ cycling taxa in the water column. In late summer, methanotrophs surpassed one million cells mL⁻¹ in FPV surface waters – a density rarely reported in any aquatic system. This bloom may reflect the metabolic advantage of Type I methanotrophs under CH₄-rich, low-O₂ conditions. While activity remains to be directly measured, these abundances suggest strong microbial capacity to intercept CH₄ before it reaches the atmosphere. This integrative approach lays essential groundwork for predicting how

microbial communities mediate greenhouse gas fluxes under engineered shading. As FPV deployment expands, incorporating microbial responses into biogeochemical assessments will be critical for managing the climate impacts of aquatic renewable energy.

Our results also demonstrate that the effect of FPV on pond biogeochemistry and CH₄ cycling varies over time. Immediately following FPV deployment in 2023, FPV ponds had lower oxygen concentrations and increased surface and bottom water CH₄ concentrations (Ray et al. 2024). By contrast, in the second year (2024), surface CH₄ concentrations no longer differed between treatments, and methanotroph abundance was substantially higher in FPV surface waters. These patterns suggest a time-lagged ecological response, where sustained hypoxia promotes bottom-water CH₄ accumulation, followed by methanotroph enrichment that buffers CH₄ flux near the surface. Quantifying how CH₄ ebullition responds to FPV deployment across seasons and years remains a key open question for understanding total emission dynamics.

Identifying strategies for low emission energy production is critical for combating climate change. As FPV expansion accelerates, understanding how this technology interacts with aquatic CH₄ dynamics is important for ensuring sustainability. Our study provides mechanistic evidence that FPV deployment can restructure microbial communities in ways that influence carbon cycling, offering a foundation for evaluating the long-term ecosystem-level impacts of continued FPV proliferation.

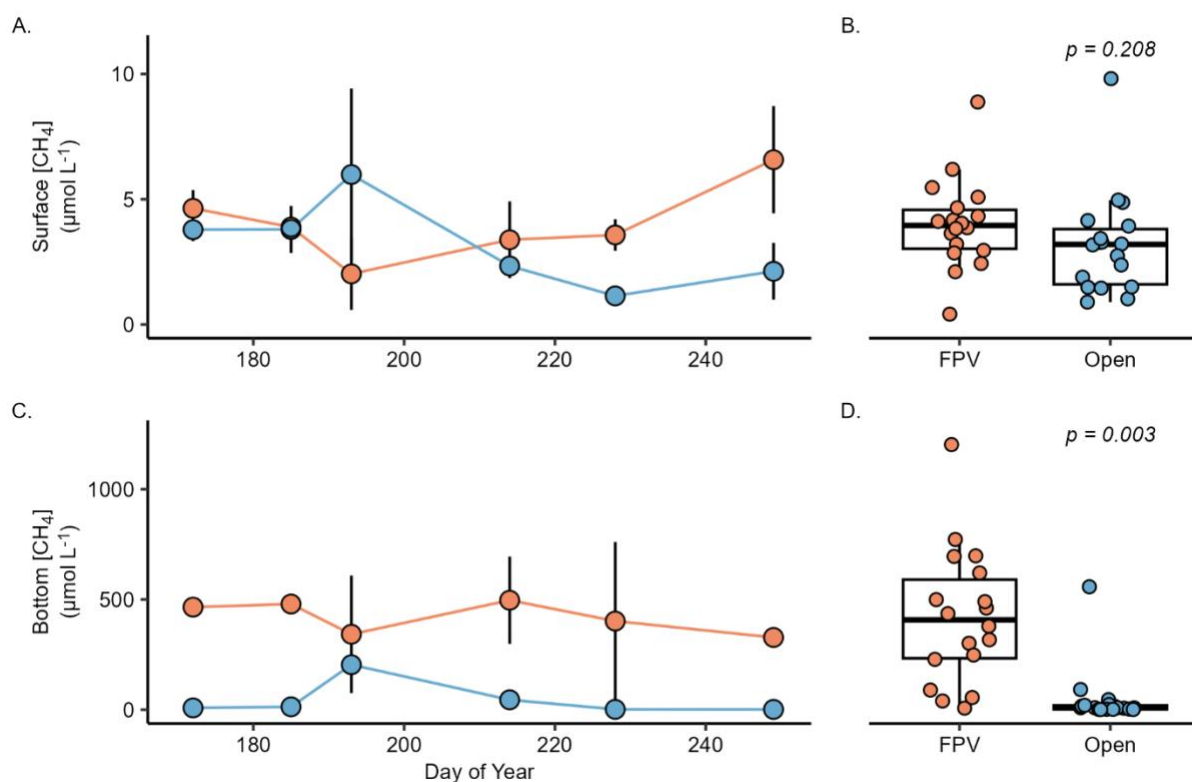


Fig. 1: Dissolved methane concentrations in the surface (A, B) and bottom water (C, D) of ponds with and without floating photovoltaic (FPV) arrays in summer 2024. In (A, C) points indicate the mean CH_4 concentration from pond centers on a given date by treatment group and error bars indicate standard deviation. Colored lines connect the means. In panels (B, D) each point on the boxplot indicates a measured CH_4 concentration in an individual pond based on whether FPV is present (FPV) or not (Open) and p-values indicate the result of a least-squares mean test of the treatment effect of the mixed model (Table S3).

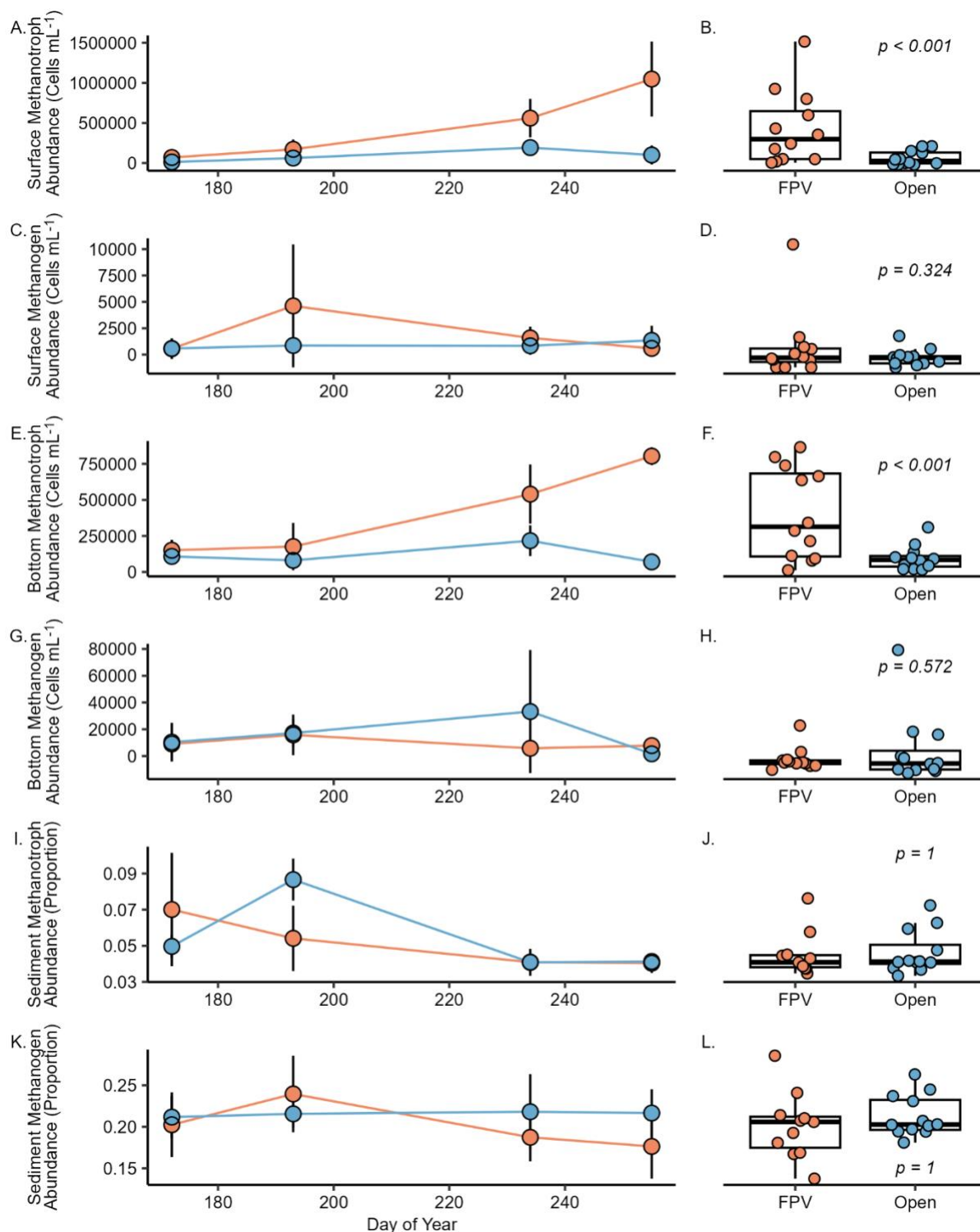


Fig. 2: Total abundance (A-H) of methanogens and methanotrophs in surface waters and bottom waters and relative abundance (I-L) of methanogens and methanotrophs in sediments of ponds with and without floating photovoltaic (FPV) arrays in summer 2024. Information as for Fig. 2.

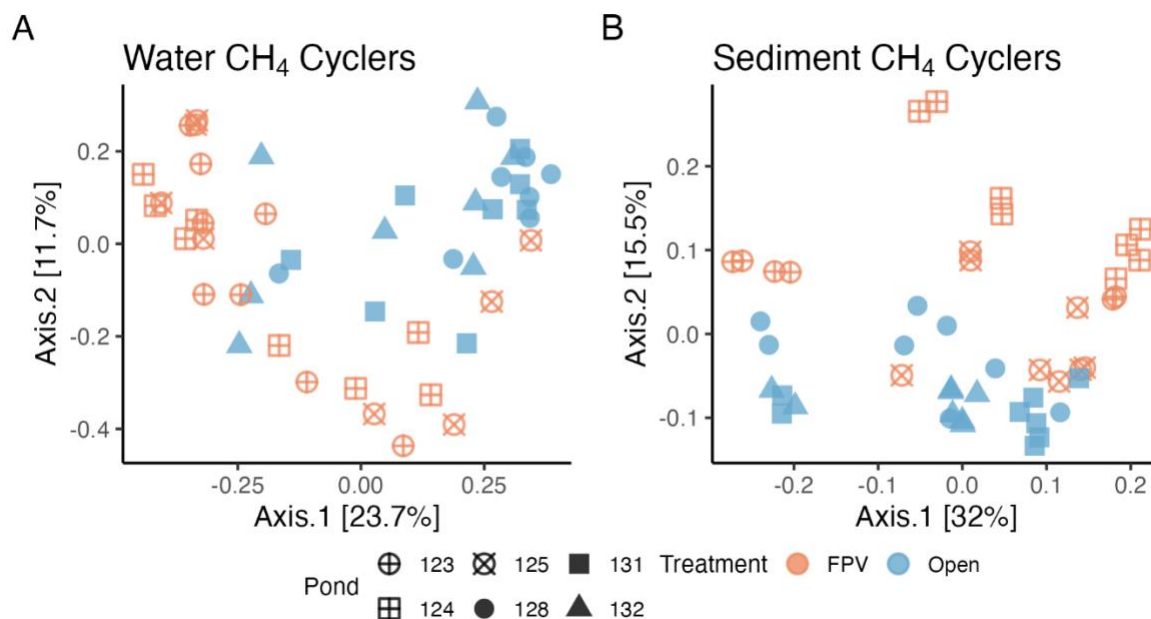


Fig. 3: Bray–Curtis PCoA of combined methanogen and methanotroph communities in ponds with floating photovoltaic arrays (“FPV”) and control ponds (“Open”). (A) Dissimilarity among water column samples was calculated using absolute cell abundances. (B) Sediment samples were rarefied to 20,822 reads to calculate dissimilarity based on relative abundance; replicates are shown separately. Shapes denote ponds. PERMANOVA and beta dispersion statistics are in Table 1.

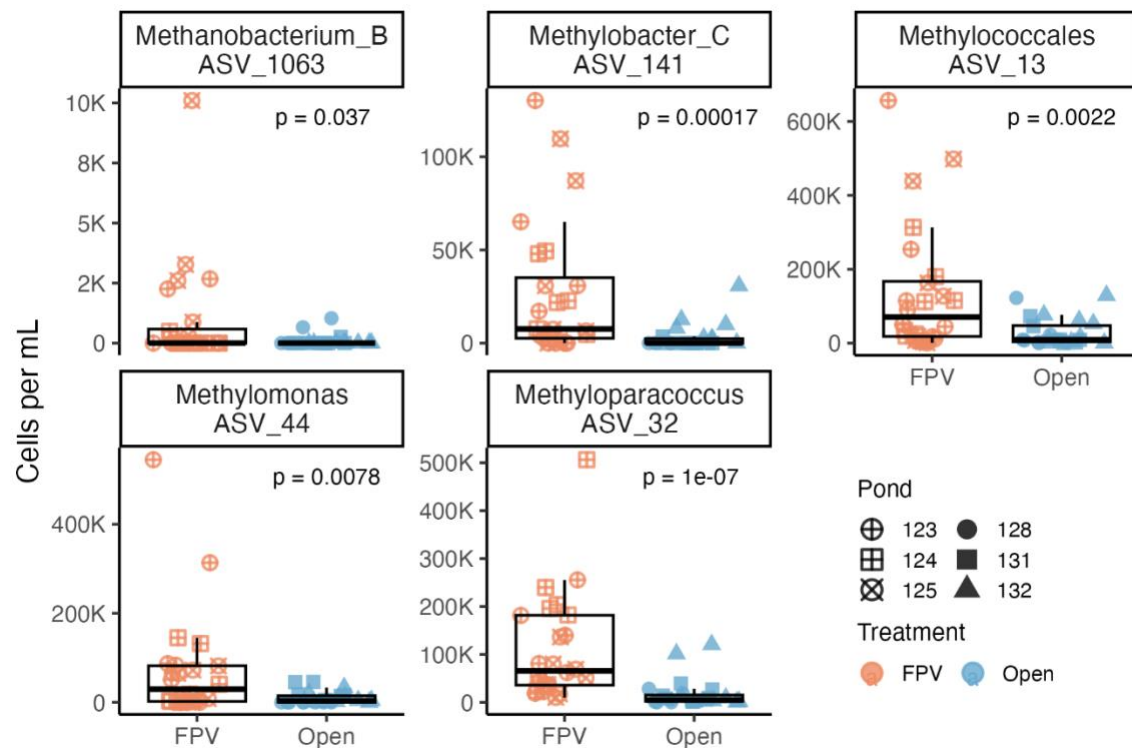


Fig. 4: Absolute cell abundances (cells mL⁻¹) of five differentially abundant ASVs identified by ANCOM-BC II in the water column of ponds with floating photovoltaic arrays (“FPV”) and control ponds without FPV (“Open”). ASV_1063 (Methanobacterium_B) is a methanogen; the remaining ASVs represent Type I methanotrophs. Shapes indicate individual ponds. All comparisons are between FPV and open ponds. P-values reflect Wilcoxon rank-sum tests.

Table 1: FPV treatment, pond, and DOY significantly influence methane-cycling community composition. PERMANOVA and β -dispersion results for CH₄-cycling communities in water and sediment samples from FPV and control ponds. Bray-Curtis dissimilarities were calculated using absolute abundances for water (Fig. 3A) and rarefied relative abundances for sediment (Fig. 3B). PERMANOVA tested effects of FPV, pond, and day of year (DOY) and their interactions; β -dispersion tested group variance. Residual R² shown for each habitat.

Habitat Type	CH ₄ Group	Variable	PERMANOVA				®-Dispersion		Figure
			R ²	Pseudo-F	p-value	Residuals	F	p-value	
Water	All	FPV	0.11381	8.2023	0.001	0.49950	7.5386	0.011	3A
Water	All	Pond	0.13354	2.4061	0.001		1.3603	0.256	3A
Water	All	DOY	0.09599	6.9181	0.001		4.8804	0.006	3A
Water	All	FPV:DOY	0.07641	5.5072	0.001				3A
Water	All	Pond:DOY	0.08075	1.4549	0.034				3A
Sediment	All	FPV	0.11101	8.4062	0.001	0.44900	0.3718	0.673	3B
Sediment	All	Pond	0.23926	4.5295	0.001		0.1485	0.835	3B
Sediment	All	DOY	0.07753	5.8706	0.001		1.7651	0.162	3B
Sediment	All	FPV:DOY	0.02773	2.0998	0.018				3B
Sediment	All	Pond:DOY	0.09548	1.8074	0.005				3B

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